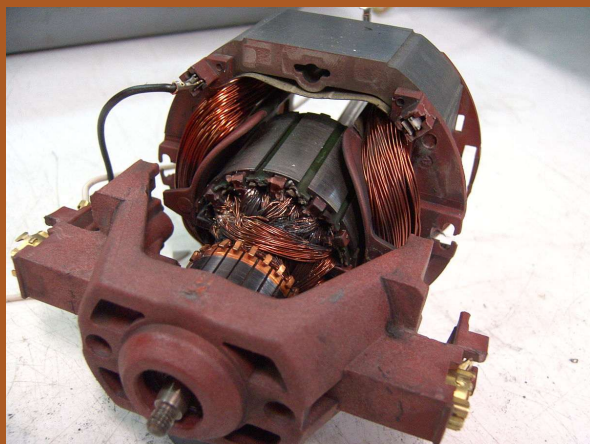


Worksheet 10

Universal motor

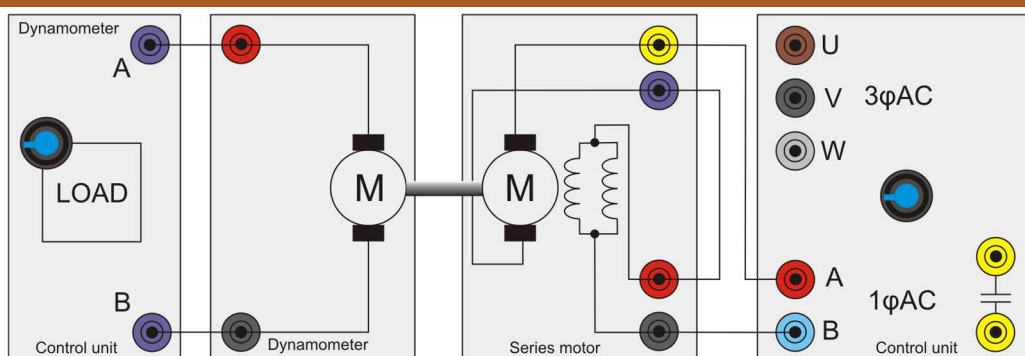
Modern electrical machines



The series-wound motor is often called the 'universal motor' because it can be operated from AC as well as DC.

This flexibility means that universal motors are amongst the most widely used.

Typical applications include planes, routers, sanders, grinders, saws etc.



This worksheet examines the AC behaviour of the series-wound motor.

Over to you:

- 1) Set up the series motor as shown above.
- 2) On your PC, load 'Open Log_1Phase.bat' and press 'Play'.
- 3) Set the Driver voltage to around 20V.
- 4) With the output frequency set at 50Hz, run the program and use the 'Sweep' function to take an automatic set of readings over a range of dynamometer loads.
- 7) When the sequence has finished, use Excel to plot a graph of speed vs torque.
- 8) Repeat this procedure for frequencies of 60, 70, and 80Hz.
- 9) On the next page sketch graphs to show traces for all frequencies.

Compare AC and DC properties of the universal motor.

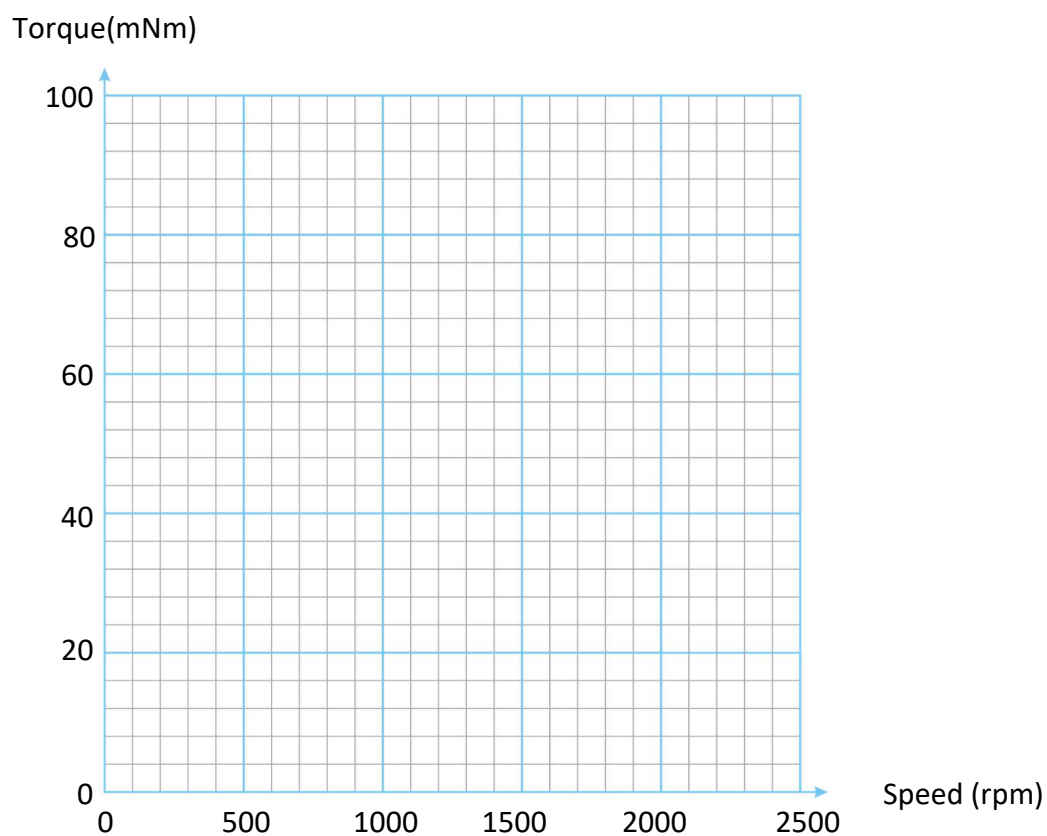
How do you change direction of motion?

Set the frequency to 40Hz - what happens?

Worksheet 11

Universal motor

Modern electrical machines



The effect of speed on torque for different values of output frequency.